



Betruck Ouma

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• EDUCATION & TRAINING

05/09/2022 - Current - NYERI, KENYA

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING- DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

- Mechanics of machines & structural mechanics (statics, dynamics, kinematics and dynamic load analysis)
- Mechanical design & modeling (3D parametric modeling, reverse engineering, CAD/CAM integration using SolidWorks and Autodesk Inventor)
- Simulation & engineering analysis (CFD, FEA, aerodynamic flow studies, motion analysis)
- Manufacturing & workshop technology (foundry operations, CNC & conventional machining, quality control)
- Applied mathematics & control systems (numerical analysis, control theory modeling, statistical DOE using MATLAB and Minitab)
- Thermodynamics & fluid mechanics (hydraulic calculations, turbomachinery design, thermal data & flow analysis)

09/09/2026

CERTIFIED SOLIDWORKS PROFESSIONAL (CSWP)- DASSAULT SYSTÈMES

01/09/2026

CERTIFIED SOLIDWORKS ASSOCIATE (CSWA)- DASSAULT SYSTÈMES

18/09/2026

CERTIFIED SOLIDWORKS SIMULATION ASSOCIATE (CSWA-S)- DASSAULT SYSTÈMES

24/09/2026

CERTIFIED SOLIDWORKS SUSTAINABILITY ASSOCIATE (CSWA-SD)- DASSAULT SYSTÈMES

27/09/2026

CERTIFIED SOLIDWORKS ADDITIVE MANUFACTURING ASSOCIATE (CSWA-AM)- DASSAULT SYSTÈMES

14/05/2024 - 13/09/2024

SOFTWARE AND WEB DEVELOPMENT - POWER LEARN PROJECT

02/09/2024 - 19/12/2024

ETHICAL HACKING- CISCO NETWORKING ACADEMY

15/01/2018 - 01/04/2022 - BUSIA, KENYA

KENYA CERTIFICATE OF SECONDARY EDUCATION (KCSE)- SIGALAME BOYS HIGH SCHOOL

• WORK EXPERIENCE

09/02/2026 - 15/05/2026 - NAIROBI, KENYA

MECHANICAL ENGINEERING ATTACHÉNUMERICAL MACHINING COMPLEX (NMC)

- CNC machining & precision manufacturing (programming and operating multi-axis CNC milling)
- Metallurgy & metal casting (pattern design allowances, metal casting, quality control, hot-dip galvanizing)
- Fabrication & structural joining (advanced thermal joining including SMAW, MIG, and TIG welding)
- Reverse engineering by CAD integration for manufacturing

27/01/2025 - 18/04/2025 - NAIROBI, KENYA

MECHANICAL ENGINEERING ATTACHÉASSOCIATED MOTORS, ISUZU

- Vehicle diagnostics & electromechanics (advanced electromechanical troubleshooting)
- Automotive servicing & maintenance (comprehensive periodic maintenance, servicing, and structural repairs on heavy-duty and light units)
- Quality assurance & safety compliance by executing Pre-Delivery Inspections [PDI]

08/01/2024 - 29/03/2024 - NYERI, KENYA

MECHANICAL ENGINEERING ATTACHÉDEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

- Mechanical design & optimization using CAD/CAM tools to improve fabrication efficiency
- Workshop operations & machining
- Installation and maintenance of dynamic mechanical systems and structural fabrication

• SKILLS

Mechanical Design | Statistical analysis (MINITAB) | CFD / FEA Simulation | MATLAB | SOLIDWORKS | AutoCAD | Ansys | Microsoft Office

• HONOURS AND AWARDS

23/02/2026 Dedan Kimathi University of Technology

Top Academic Performer Recognized as the best student in the Bachelor of Science in Mechanical Engineering program for three consecutive academic years (1st Year, 2nd Year, and 3rd Year).

• PROJECTS

03/11/2025 - 04/02/2026

10kW Crossflow Turbine design Performed complete hydraulic dimensioning and structural design of a 10 kW crossflow turbine for deployment on Kenyan rivers, sizing the runner geometry, nozzle and casing.

31/08/2025 - 19/11/2025

Biogas Generator Conversion Converted a gasoline engine into a 4 kW biogas-fueled power generator, redesigning fuel delivery, compression ratio, and combustion chamber.

19/03/2026 - 05/05/2026

Re-Block Extruder Proposed an extruder machine converting recycled plastics and sand into paving blocks, targeting plastic waste reduction and low cost construction materials.

15/05/2024 - 01/08/2024

Maize Cob Crusher & Compactor Designed a maize cob crusher and compactor to process maize crop residue into a densified output, sizing both the crushing and compaction mechanisms for required throughput.

24/02/2025 - 01/04/2025

Speed Reducer Gearbox Designed a compound reverted helical gearbox (78.5 kW, 1200-155 rpm) via AGMA gear-stress and shaft fatigue analysis, achieving factor of safety (FOS) > 2 across all shafts.